

CTMT Dynamics II

The Irreversibility Lock, the Gaussian Ceiling, and a Ranked Road from Linear to Physical Dynamics

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Abstract

This is a working map of the passage from CTMT’s linear-stochastic dynamics to genuine physics: what can be proved now, and what must still be developed. Two results are *locked*. First (the irreversibility lock), the stationary entropy production of the linear layer is $\sigma_{\text{EP}} = \text{tr}(\nu^\top D^{-1} \nu \Sigma)$ with $\nu = A + D\Sigma^{-1}$; it vanishes exactly at detailed balance ($A\Sigma$ symmetric), and for isotropic noise the sole source of irreversibility is the *antisymmetric* part of the drift—in particular the antisymmetric part of the resolved–null cross-drift. This gives CTMT its first genuinely physical output: a thermodynamic arrow of time sourced by sector circulation. Second (the Gaussian ceiling), the linear-additive layer is Gaussian-closed—all cumulants above second vanish and the moment hierarchy closes—so no higher-order “complexity gain” can arise within it; any nonlinear drift or multiplicative noise is the precise entry point of non-Gaussian structure. Both are numerically verified. The remainder is a roadmap of open targets, ranked from the largest physical payoff if proven (nonlinear non-equilibrium transport) to the smallest and most attackable (information-causal ordering), each with its tool and its honest failure mode. We also retire, with reasons, the Fisher-induced Lorentz-signature construction of the earlier causality draft, keeping only its correct residue. The honest conclusion: the physics CTMT can actually reach is *irreversible transport thermodynamics*, and the arrow of time is already in hand.

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1 How to read this paper

This document separates three registers, and labels every statement by which one it belongs to:

Locked.

Proved here from standard mathematics and numerically verified (Sections 3–4). These will not move.

Roadmap.

Precisely posed open targets (Section 5), ranked by physical payoff, each with the tool that should settle it and the way it could fail.

Retired.

Earlier physical claims that cannot be locked, withdrawn with reasons (Section 6).

The strategic point up front: the linear-stochastic layer has exactly one edge that touches real physics without overreach—*irreversible thermodynamics*. That edge is now a theorem. Everything grander (nonlinearity, spacetime, causality-as-Lorentz) is either a ranked target or retired.

2 Setup

The dynamical layer (Dynamics I) is the linear-stochastic law

$$dy = A y dt + B dW, \quad Q := BB^\top \succ 0, \quad D := \tfrac{1}{2}Q,$$

on the observation space $O = R \oplus N$ (resolved/null, g -orthogonal). With A Hurwitz the stationary covariance Σ solves the algebraic Lyapunov equation $A\Sigma + \Sigma A^\top + Q = 0$. In the adapted frame the drift has blocks A_{RR}, A_{NN} (within-sector) and A_{RN}, A_{NR} (cross-sector); the cross-drift is the dynamical form of “seepage” (Dynamics I: it generates the coupling and constitutes the Mori–Zwanzig memory kernel $A_{RN}e^{A_{NN}\tau}A_{NR}$).

3 Lock 1 — the irreversibility theorem

Theorem 3.1 (Entropy production and the source of irreversibility). *For the linear-stochastic layer above:*

1. The stationary (non-equilibrium steady state) entropy production rate is

$$\sigma_{\text{EP}} = \text{tr}(\nu^\top D^{-1} \nu \Sigma) \geq 0, \quad \nu := A + D\Sigma^{-1}.$$

2. $\sigma_{\text{EP}} = 0 \iff \nu = 0 \iff A\Sigma = -D$ is symmetric (detailed balance).
3. (Isotropic lemma.) If $Q = I$ then detailed balance holds iff A is symmetric, and then $\Sigma = -\frac{1}{2}A^{-1}$.
4. (Sector source.) With $Q = I$ and symmetric within-sector blocks, the cross-drift contributes to σ_{EP} only through its antisymmetric part $A_{NR} - A_{RN}^\top$: symmetric coupling ($A_{NR} = A_{RN}^\top$) preserves detailed balance, while antisymmetric (circulating) coupling strictly sources irreversibility.

Proof. (1) The stationary density is $\rho \propto e^{-\frac{1}{2}y^\top \Sigma^{-1}y}$ and the Fokker–Planck probability current is $J = Ay\rho - D\nabla\rho = (A + D\Sigma^{-1})y\rho = \nu y\rho$. The mean entropy production rate of a linear diffusion is $\sigma_{\text{EP}} = \int (J^\top D^{-1}J)/\rho dy = \int (\nu y)^\top D^{-1}(\nu y)\rho dy = \text{tr}(\nu^\top D^{-1}\nu\Sigma)$, using $\int yy^\top \rho dy = \Sigma$ [1, 2]. (2) $\nu = 0 \iff A = -D\Sigma^{-1} \iff A\Sigma = -D$; symmetry of $-D$ is the detailed-balance condition. (3) If $Q = I$ and $A = A^\top \prec 0$ then $\Sigma = -\frac{1}{2}A^{-1}$ solves $A\Sigma + \Sigma A^\top + I = 0$ ($-\frac{1}{2}I - \frac{1}{2}I + I = 0$) and $A\Sigma = -\frac{1}{2}I$ is symmetric, so detailed balance holds; conversely

detailed balance forces $A\Sigma = -\frac{1}{2}I$, hence $A = -\frac{1}{2}\Sigma^{-1}$ is symmetric. (4) With $Q = I$ and symmetric A_{RR}, A_{NN} , the matrix A is symmetric iff $A_{NR} = A_{RN}^\top$; by (3) A symmetric $\iff$ detailed balance $\iff \sigma_{\text{EP}} = 0$. Thus $\sigma_{\text{EP}} > 0$ iff the antisymmetric cross part is nonzero. Verified numerically: symmetric coupling gives $\sigma_{\text{EP}} = 0$ to machine precision at all strengths, antisymmetric coupling gives σ_{EP} growing with strength. $\square$

Remark 3.2 (The physical content: a thermodynamic arrow). σ_{EP} is a genuine physical quantity—the rate of entropy production of an irreversible process—not an information-geometric analogy. Its positivity is a second-law statement, and it defines a strictly monotone thermodynamic time (cumulative entropy production) precisely when the dynamics is irreversible ($\nu \neq 0$). CTMT’s seepage, in its circulating (antisymmetric) component, is the microscopic source of this arrow. This is the first place the framework outputs physics rather than estimation geometry.

Remark 3.3 (Correcting the “coherence proper time” instinct). The earlier causality draft proposed a time $\tau(t) = \int_0^t \sqrt{\lambda_{\max}(F)} dt'$. This is monotone, but only trivially (the integrand is nonnegative); it is a clock *reparametrization* and encodes no irreversibility—a fully reversible system has the same τ . The physically meaningful monotone arrow is cumulative entropy production, which increases iff the process is genuinely irreversible. The honest realization of the “Fisher→time” idea is thermodynamic (Theorem 3.1), not Lorentzian (see Section 6).

4 Lock 2 — the Gaussian-closure ceiling

Theorem 4.1 (Gaussian closure and the entry point of complexity). *Let $dy = Ay dt + B dW$ with Gaussian or deterministic initial condition.*

1. $y(t)$ is Gaussian for all t ; the stationary law is $\mathcal{N}(0, \Sigma)$; all cumulants of order ≥ 3 vanish.
2. The moment hierarchy closes at second order: the covariance obeys the Lyapunov equation with no coupling to third or higher moments.

Consequently, within the linear-additive class no non-Gaussian structure (skewness, kurtosis, and a fortiori chaos, intermittency, or pattern formation) can be generated. If the drift is nonlinear or the noise is multiplicative ($\sigma(y)$), the hierarchy no longer closes and higher cumulants appear: nonlinearity is the precise entry point of complexity.

Proof. (1) The solution $y(t) = e^{At}y(0) + \int_0^t e^{A(t-s)}B dW(s)$ is a linear functional of jointly Gaussian inputs, hence Gaussian; its cumulants above second vanish. (2) $\frac{d}{dt}\mathbb{E}[yy^\top] = A\mathbb{E}[yy^\top] + \mathbb{E}[yy^\top]A^\top + Q$ involves only the second moment. For nonlinear drift $b(y)$, $\frac{d}{dt}\mathbb{E}[yy^\top]$ contains $\mathbb{E}[y b(y)^\top]$, which for b with nonzero curvature couples to third and higher moments, breaking closure; multiplicative noise does likewise through $\mathbb{E}[\sigma(y)\sigma(y)^\top]$. Verified numerically: a purely linear drift yields stationary excess kurtosis ≈ 0 , while adding a cubic drift produces clearly nonzero excess kurtosis. $\square$

Corollary 4.2 (Why physics is not reachable from monotone linear math). *The intuition that “complexity gain is not reachable from simple linear dynamics” is a theorem, not a short-fall: the linear layer is Gaussian-closed (Theorem 4.1) and, generically, near-reversible with computable entropy production (Theorem 3.1). Genuine complexity requires crossing into the nonlinear/multiplicative class. Proving the ceiling is therefore valuable in itself: it delimits the linear layer exactly and names the crossing.*

Lemma 4.3 (Static resolution geometry is flat; curvature lives in the family). *For fixed J, Σ the resolved response metric $J_R^\top S_R^{-1} J_R$ is constant, so a single chart carries no intrinsic (Fisher) curvature. Curvature is a property of the family $\{\mathcal{G}(\theta)\}$ as the operating point varies: it is the curvature of the projection connection on the moving resolved subbundle (Dynamics I / Atlas), governed by Ambrose–Singer.*

Proof. A constant positive-definite form on a vector space is flat. Variation of the metric requires variation of (J, Σ) over a base, i.e. the bundle setting. $\square$

This is why “curvature” cannot be read off a single CTMT chart; it is a roadmap item requiring the moving-chart structure (target R4 below).

5 Roadmap, ranked by physical payoff

Targets are ordered from the largest physical gain if proven to the smallest, with the tool and the honest failure mode for each.

Roadmap target 5.1 (Nonlinear non-equilibrium transport — *highest payoff*). **Goal.** Extend Lock 1 to nonlinear diffusions: derive entropy production, a fluctuation theorem (Crooks/Jarzynski type), and transport coefficients (Green–Kubo, from the memory kernel) for the CTMT resolved sector. **If proven**, CTMT becomes a genuine framework for irreversible transport physics — the strongest possible outcome short of fundamental physics. **Tool.** Stochastic thermodynamics, Green–Kubo relations, weak-noise large deviations. **Failure mode / attack.** The nonlinear Mori–Zwanzig kernel does not close in general; unqualified claims will be attacked on closure. *Strategy:* start with weak nonlinearity (perturbative) or specific solvable classes; prove the fluctuation theorem there first.

Roadmap target 5.2 (Fluctuation–dissipation cross-exponent test — empirical physics). **Goal.** Derive the relation the linear model *forces* between the memory-kernel decay exponent and an independently measured fluctuation exponent, then test it on real relaxation data. **If proven and confirmed**, this is genuine empirical support (or a clean refutation). **Tool.** Fluctuation–dissipation theorem; time-series estimation. **Failure mode / attack.** Must relate *independently* measured exponents; fitting one power law and reading it back is reparametrization, not confirmation (cf. the Omori consistency result).

Roadmap target 5.3 (Coarse-grained second law for the resolved sector). **Goal.** On eliminating the null sector (Mori–Zwanzig), derive the resolved-sector entropy-production inequality and the information cost of coarse-graining (Landauer / Sagawa–Ueda). **If proven**, ties seepage to the thermodynamics of information and measurement. **Tool.** Stochastic thermodynamics of coarse-graining. **Reachability.** Closest to lockable after Lock 1. **Failure mode / attack.** Apparent vs. total entropy production must be separated carefully, or the inequality direction can be mis-stated.

Roadmap target 5.4 (Curvature compatibility (information-geometric vs bundle)). **Goal.** Show the Amari information-geometric curvature of the family and the projection-bundle curvature (Lemma 4.3) agree under a stated compatibility condition (Dynamics I, Rungs 4–5). **If proven**, yields an intrinsic geometric invariant. **Tool.** Amari α -connections; Ambrose–Singer. **Failure mode / attack.** This is *statistical/model* curvature, not spacetime curvature; presenting it as gravity is the attack that sank earlier drafts. Keep it geometric.

Roadmap target 5.5 (Information-causal ordering — *lowest payoff, most attackable*). **Goal.** Make the coherence causal-ordering functional rigorous as an *ordering*, not a metric. **If proven**, gives an information ordering of events. **Tool.** Statistical arc length; directed information. **Failure mode / attack.** The “Fisher-induced Lorentz signature” is not derivable (Section 6); any claim of emergent spacetime or Lorentz invariance will be rejected. Restrict to an ordering functional and claim nothing metric.

6 Retired, with reasons

Non-claim 6.1 (Fisher-induced Lorentz signature). The causality draft’s line element $d\tau^2 = \lambda_{\max}^{-1} dt^2 - \lambda_{\min}^{-1} d\ell^2$ obtains its hyperbolic signature from the *choice of sign* on the two terms; the signature is inserted, not derived. Fisher information is positive semidefinite, so nothing in it forces a negative eigenvalue. This construction is withdrawn as a derivation of Lorentz structure. (An honest hyperbolic signature can appear in a rate–distortion Hessian only after *assuming* distortion penalizes temporal mis-ordering more than spatial dispersion—an input, not an output.)

Non-claim 6.2 (Mesoscopic-quantum and biological anchorings). The predicted pre-decoherence deviations and cancer-progression coherence breakdowns are speculative pattern-matching without a derived, quantitative, falsifiable relation. They are withdrawn from the claimed core and, if pursued, belong under target R2 (a forced, independently-testable relation) with real data.

Remark 6.3 (Salvaged residue). Two correct pieces of the causality/physics drafts survive. *Redundancy = Fisher null space*: directions in $\ker(J^\top \Sigma^{-1} J)$ are exactly the null sector, propagating no distinguishability—this is already the CTMT resolved/null decomposition, correctly stated. And *arc-length as a weak clock*: $\tau = \int \sqrt{\lambda_{\max}} dt$ is a valid monotone reparametrization, now understood as physically inert and superseded by the entropy-production arrow of Lock 1.

7 Conclusion

Two things are now fixed. The linear-stochastic layer has a genuine physical output—its stationary entropy production, sourced by the antisymmetric part of the resolved–null circulation—so CTMT possesses a thermodynamic arrow of time that is a theorem, not an analogy. And that same layer has a proved ceiling: being Gaussian-closed, it cannot manufacture complexity, which enters only with nonlinearity. Together these say precisely what the framework can and cannot do in its current, defensible form.

The ranked roadmap then shows where the physical payoff is. The reachable finish line is not spacetime, turbulence, or Lorentz causality; it is *irreversible transport thermodynamics*—non-equilibrium entropy production, fluctuation theorems, transport coefficients—extended from the linear lock into the nonlinear regime. That is the domain the twenty-year through-line, from a delayed wall-tap to a memory kernel to entropy production, was always pointing at. The arrow of time is in hand; the nonlinear thermodynamics is the road worth walking, and the Lorentzian and spacetime readings are honestly set down.

References

- [1] T. Tomé and M. J. de Oliveira, “Entropy production in irreversible systems described by a Fokker–Planck equation,” *Phys. Rev. E* 82 (2010), 021120.
- [2] U. Seifert, “Stochastic thermodynamics, fluctuation theorems and molecular machines,” *Rep. Prog. Phys.* 75 (2012), 126001.
- [3] C. Gardiner, *Stochastic Methods*, 4th ed., Springer, 2009.
- [4] R. Zwanzig, “Memory effects in irreversible thermodynamics,” *Phys. Rev.* 124 (1961), 983–992.
- [5] R. Kubo, “The fluctuation–dissipation theorem,” *Rep. Prog. Phys.* 29 (1966), 255–284.
- [6] S. Amari, *Information Geometry and Its Applications*, Springer, 2016.

- [7] T. Sagawa and M. Ueda, “Generalized Jarzynski equality under nonequilibrium feedback control,” *Phys. Rev. Lett.* 104 (2010), 090602.